

1 General questions

(a) What is the core difference between an MDP problem and an RL problem?

Solution:

An MDP is specified in terms of a tuple $\langle S, A, T, R \rangle$, where the state transition matrix $T(s, a, s')$ and the reward function $R(s, a, s')$ represent the known *model* of the world in terms of effects of actions (i.e., transition probabilities from one state to another after executing an action) and of expected immediate rewards. The values of the states and of the q-states satisfy the Markov property and, accordingly, the Bellman's equation. The goal is to find (i.e., compute) a (deterministic) policy π that maximize the discounted sum of the gathered rewards (or minimize the sum of payed costs) over a potentially infinite life horizon.

In an RL problem scenario, the setting and the goal are the same but the agent doesn't have the model of the world such that it has to *learn* through a trial and error experience how the world behave and learn the optimal policy accordingly.

Note the difference between computing an optimal policy and learning an optimal policy. In an MDP since everything which is needed is given, the problem boils down to perform an efficient computation of the optimal policy (e.g., via value iteration and its variants, or via policy iteration). In RL, the agent has to try out states and actions and observe actual transitions and rewards in order to learn a policy. More precisely, in Passive RL, first, T and R are estimated from experience, and then the resulting approximate MDP is used to compute the policy. In Active RL, the policy (i.e., the Q function) is estimated directly, without passing through building a world model first.

(b) What is the difference between model-based and model-free learning?

Solution: For model-based learning, the agent learns an approximate model of the world (transitions and rewards) based on experiences and solves for values (e.g., using value or policy iteration) as if the learned model were correct (i.e., it assumes that the estimated MDP is correct). A model-free learning is an algorithm which does not builds or directly use a model in terms of the transition probability distribution or reward function.

2 Reinforcement Learning

Imagine an unknown environments with four states (A, B, C, and X), two actions ($\leftarrow$ and $\rightarrow$). An agent acting in this environment has recorded the following episode:

s	a	s'	r	Q-learning iteration numbers (for part b)
A	$\rightarrow$	B	0	1, 10, 19, ...
B	$\rightarrow$	C	0	2, 11, 20, ...
C	$\leftarrow$	B	0	3, 12, 21, ...
B	$\leftarrow$	A	0	4, 13, 22, ...
A	$\rightarrow$	B	0	5, 14, 23, ...
B	$\rightarrow$	A	0	6, 15, 24, ...
A	$\rightarrow$	B	0	7, 16, 25, ...
B	$\rightarrow$	C	0	8, 17, 26, ...
C	$\rightarrow$	X	1	9, 18, 27, ...

- (a) Consider running model-based reinforcement learning based on the episode above. Calculate the following quantities:

$$\hat{T}(B, \rightarrow, C) = \frac{2}{3}$$

$$\hat{R}(C, \rightarrow, X) = 1$$

- (b) Now consider running Q-learning, repeating the above series of transitions in an infinite sequence. Each transition is seen at multiple iterations of Q-learning, with iteration numbers shown in the table above.

After which iteration of Q-learning do the following quantities first become nonzero? (If they always remain zero, write *never*).

$$Q(A, \rightarrow)? \quad 14$$

$$Q(B, \leftarrow)? \quad 22$$

- (c) True/False: For each question, you will get positive points for correct answers, zero for blanks, and negative points for incorrect answers. Circle your answer **clearly**, or it will be considered incorrect.

- (i) [true or false] In Q-learning, you do not learn the model.

Q learning is model-free, you learn the optimal policy explicitly, and the model itself implicitly.

- (ii) [true or false] For TD Learning, if I multiply all the rewards in my update by some nonzero scalar p , the algorithm is still guaranteed to find the optimal policy.

TD Learning does not necessarily find the optimal policy, it only learns the value of the states following some given policy.

- (iii) [true or false] In Direct Evaluation, you recalculate state values after each transition you experience.

In order to estimate state values, you calculate state values from episodes of training, not single transitions.

- (iv) [true or false] Q-learning requires that all samples must be from the optimal policy to find optimal q-values.

Q-learning is off-policy, you can still learn the optimal values even if you act suboptimally sometimes.